CORE PURE / POLAR COORDINATES

# Polar Coordinates Basics

Mark scheme 10 questions · 97 marks

EXAM BOARDS Edexcel 9FM0 · AQA 7367 · OCR A H245 · OCR MEI H645 · CIE 9231

## Questions


| Question | Marks by part | Marks | Page |
|---|---|---|---|
| Question 1 | (a) 3 (b) 2 (c) 3 | 8 | 4 |
| Question 2 | (a) 1 (b) 3 (c) 4 | 8 | 5 |
| Question 3 | (a) 1 (b) 6 | 7 | 6 |
| Question 4 | (a) 4 (b) 4 | 8 | 8 |
| Question 5 | (a) 4 (b) 4 (c) 3 | 11 | 9 |
| Question 6 | (i) 4 (ii) 3 | 7 | 11 |
| Question 7 | (a) 3 (b) 3 (c) 2 (d) 4 | 12 | 12 |
| Question 8 | (i) 2 (ii)(a) 4 (ii)(b) 3 (iii) 4 | 13 | 14 |
| Question 9 | (a) 2 (b) 6 | 8 | 16 |
| Question 10 | (a) 3 (b)(i) 3 (b)(ii) 5 (b)(iii) 4 | 15 | 17 |
| | **Total** | **97** | |

# Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

### Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

| Code | Meaning |
|---|---|
| **M1** | Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted. |
| **dM1** | Dependent method: as M1, but only if the method mark before it was earned, unless its note names another. |
| **A1** | Accuracy: a correct answer or step. It needs the method mark it follows. |
| **A1ft** | As A1, following through from an earlier answer that the note names. |
| **A1 AG** | The last mark of a part whose answer is printed in the question: complete working, with no errors. |
| **B1** | A correct value, fact or statement. It needs no method mark. |
| **B1ft** | As B1, following through from an earlier answer that the note names. |
| **E1** | A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark. |

### Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes. A grey line gives the rule where the name alone doesn't.

| | | | |
|---|---|---|---|
| **ft** | follow through<br>Correct work from the student's own earlier wrong value earns the mark. | **AG** | answer given<br>The answer is printed in the question, so every step to it must be shown. |
| **cao** | correct answer only<br>No follow-through from an earlier error. | **cso** | correct solution only<br>No error anywhere in the part, not just in its answer. |
| **oe** | or equivalent | **awrt** | answers which round to |
| **soi** | seen or implied<br>Earned if written, or if later working shows it was used. | **isw** | ignore subsequent working<br>A correct answer, once reached, isn't spoiled by later working. |
| **SC** | special case<br>Credit, as the note says, for work outside the main method. | **their** | the student's own earlier value |
| **condone** | a slip that doesn't cost the mark | [ ] | working the mark doesn't need |
| **s.f.** | significant figures | **d.p.** | decimal places |

**Reading the scheme**

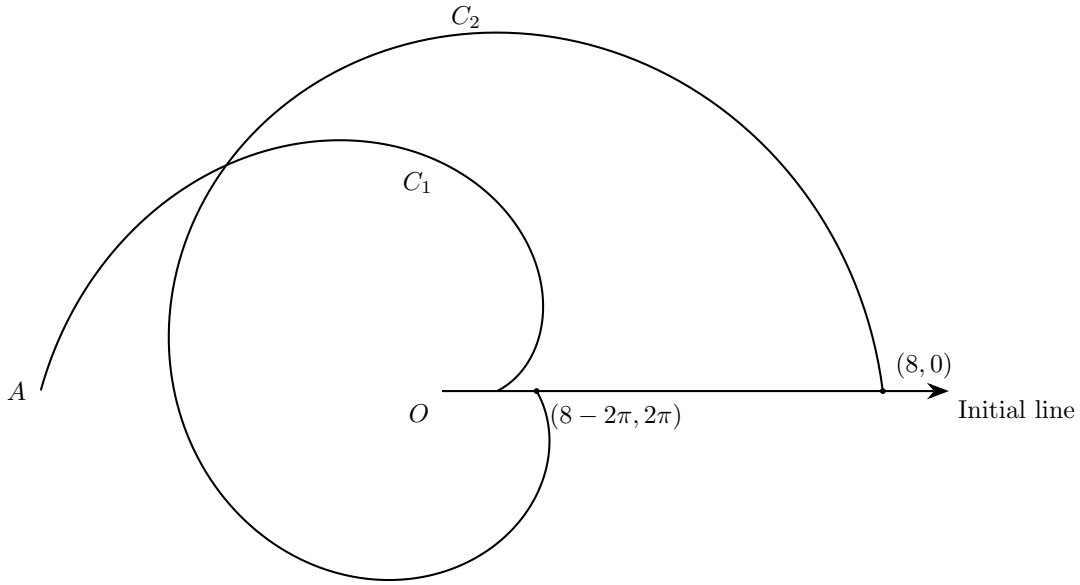
- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- [***n***] ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

**Accuracy and answers**

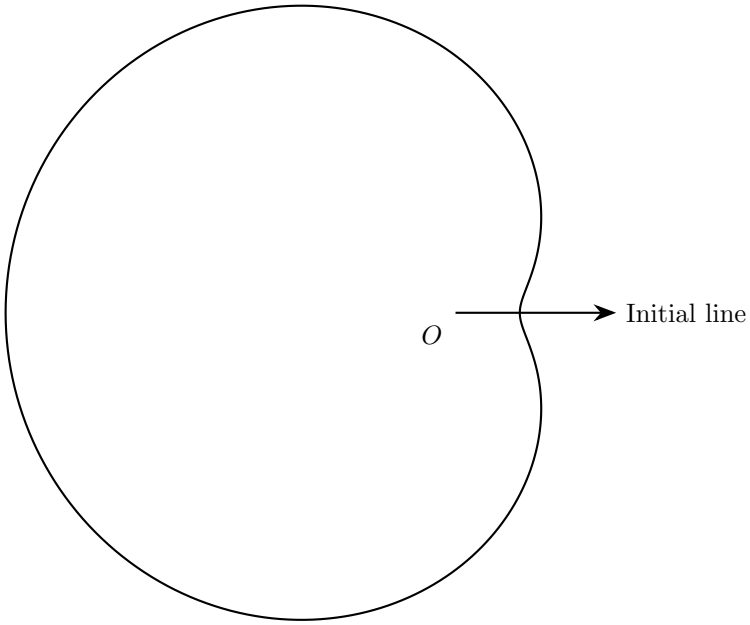
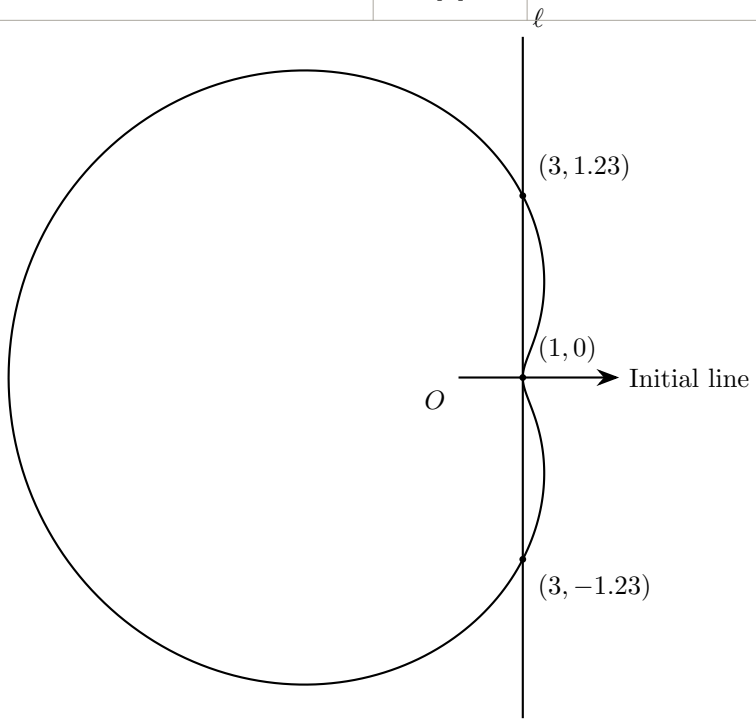
- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

**Value of $g$**

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state $g$, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

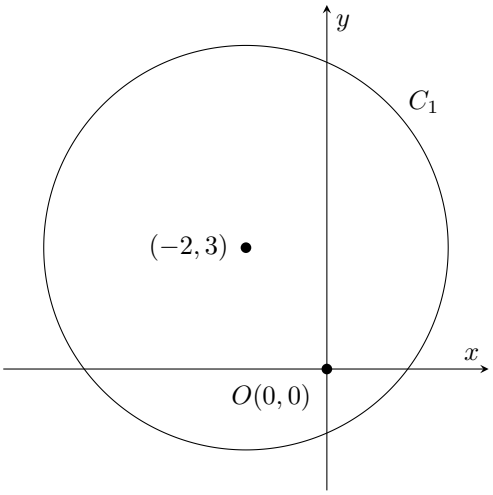
| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **1(a)** | $A = (1 + 2\pi, \pi)$ | **B1** | Accept $r$ awrt 7.28; $\theta = \pi$ or $180°$. |
| | $x =$ "$(1 + 2\pi)$" $\cos\pi$, $y =$ "$(1 + 2\pi)$" $\sin\pi$ | **M1** | Uses $x = r\cos\theta$ and $y = r\sin\theta$ with their $r$ and $\theta = \pi$. |
| | $(-1 - 2\pi, 0)$ | **A1** | oe, or awrt $(-7.28, 0)$. |
| | | **[3]** | |
| **1(b)** | $1 + 2\theta = 8 - \theta \Rightarrow \theta = \ldots$ | **M1** | Equates the two expressions for $r$ and solves for $\theta$. Condone sign slips. |
| | $\left(\frac{17}{3}, \frac{7}{3}\right)$ | **A1** | oe, or awrt $(5.67, 2.33)$. Both coordinates needed. |
| | | **[2]** | |
| **1(c)** |  | | |
| | *Shape* | **B1** | Spiral making one anticlockwise turn with $r$ decreasing throughout, starting and ending on the initial line, not reaching $O$. |
| | *Labelled point* | **B1** | Start on the initial line labelled 8 or $(8, 0)$. |
| | *Labelled point* | **B1** | End on the initial line between $O$ and the start, labelled $8 - 2\pi$, awrt 1.72, or $(8 - 2\pi, 2\pi)$. |
| | | **[3]** | |
| | **Total for Question 1** | **8** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **2(a)** | $r = \dfrac{6}{2-\sin\frac{\pi}{6}} = \dfrac{6}{2-\frac{1}{2}} = 4$ | **B1** | Needs $\sin\frac{\pi}{6} = \frac{1}{2}$ substituted, or the denominator $\frac{3}{2}$ seen, leading to $r = 4$. |
| | | **[1]** | |
| **2(b)** | $r$ greatest when $2 - \sin\theta$ is least: $\sin\theta = 1$ | **B1** | Identifies $\sin\theta = 1$, or $\theta = \frac{\pi}{2}$, as giving greatest $r$, by the least denominator or otherwise. |
| | $\theta = \frac{\pi}{2} \Rightarrow r = \dfrac{6}{2-1} = \ldots$ | **M1** | Substitutes their $\theta$, from maximising $r$, into the equation of $C$ to find $r$. |
| | $\left(6, \frac{\pi}{2}\right)$ | **A1** | Accept $r = 6$, $\theta = \frac{\pi}{2}$ stated separately. |
| | | **[3]** | |
| **2(c)** | $r\sin\theta = r\cos\theta \Rightarrow \tan\theta = 1$ | **M1** | Substitutes $x = r\cos\theta$, $y = r\sin\theta$ into $y = x$, reaching $\tan\theta = 1$ or an angle directly. |
| | $\theta = \frac{\pi}{4},\ -\frac{3\pi}{4}$ | **A1** | Both, with no others in $-\pi < \theta \leq \pi$. Condone $\frac{5\pi}{4}$ for the second. |
| | $r = \dfrac{6}{2-\frac{\sqrt{2}}{2}},\ x = r\cos\frac{\pi}{4} = \ldots$<br>$r = \dfrac{6}{2+\frac{\sqrt{2}}{2}},\ x = r\cos\left(-\frac{3\pi}{4}\right) = \ldots$ | **M1** | Finds $r$ at one of their angles and uses $x = r\cos\theta$ or $y = r\sin\theta$. Need not be simplified. |
| | $\left(\frac{6+12\sqrt{2}}{7}, \frac{6+12\sqrt{2}}{7}\right), \left(\frac{6-12\sqrt{2}}{7}, \frac{6-12\sqrt{2}}{7}\right)$ | **A1** | Both points, each with both coordinates, in any exact form; need not be rationalised. |
| | | **[4]** | |
| **Total for Question 2** | | **8** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **3(a)** |  | | |
| | *Symmetry* | **B1** | Given portion reflected in the initial line, giving a closed curve symmetric about it and still passing through $(1, 0)$. |
| | | **[1]** | |
| **3(b)** |  | | |
| | *Intercepts* | **B1** | Vertical line through $(1, 0)$ on the initial line, meeting $C$ there and crossing it once above and once below the initial line. |

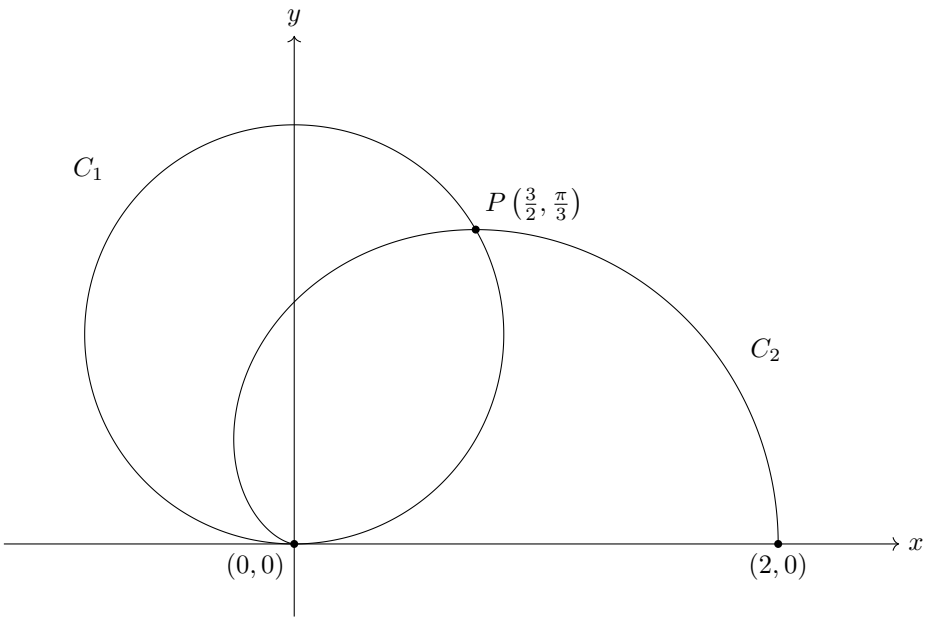
*Question 3 continued*

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| | $(4-3\cos\theta)\cos\theta = 1$ | **M1** | Substitutes for $r$ in $r\cos\theta = 1$ to obtain an equation in $\cos\theta$ alone. |
| | $3\cos^2\theta - 4\cos\theta + 1 = 0$ | **A1** | oe, any three-term form. |
| | $(3\cos\theta - 1)(\cos\theta - 1) = 0 \Rightarrow \cos\theta = \ldots$<br>$\theta = \ldots,\ r = 4 - 3\cos\theta = \ldots$ | **dM1** | Solves their three-term quadratic in $\cos\theta$ by any means, then finds at least one $\theta$ and its $r$ from either equation. |
| | $(1, 0)$ | **A1** | |
| | $(3, 1.23),\ (3, -1.23)$ | **A1** | Both, with $r = 3$ and angles awrt $\pm 1.23$. A0 for an angle outside $-\pi < \theta \leq \pi$. |
| | | **[6]** | |
| | **Total for Question 3** | **7** | |

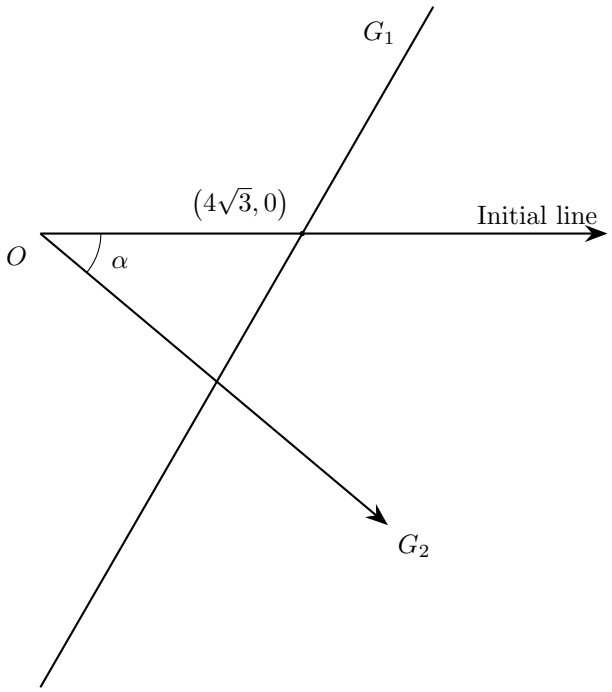
| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **4(a)** | $x^2 + y^2 + 4x - 6y = 12$ | **M1 A1** | **M1** Replaces $r^2$ by $x^2 + y^2$ and at least one of $r\cos\theta$, $r\sin\theta$ by $x$, $y$.<br>**A1** Correct equation, oe. |
| | $(x+2)^2 - 4 + (y-3)^2 - 9 = 12$ | **M1** | Completes the square in both $x$ and $y$ for their equation of the form $x^2 + y^2 + ax + by = c$. Condone sign slips. |
| | $(x+2)^2 + (y-3)^2 = 25$<br>Centre $(-2, 3)$, radius 5 | **A1** | Both centre and radius, from a correct equation in circle form. |
| | | **[4]** | |
| **4(b)** |  | | |
| | *Centre and radius* | **B1** | Circle with its centre marked in the second quadrant, labelled $(-2, 3)$. |
| | *Labelled point* | **B1** | Pole marked at the origin, inside the circle. |
| | $OA = \sqrt{(-2)^2 + 3^2}$ $[= \sqrt{13}]$<br>$5 \pm$ "$\sqrt{13}$" | **M1** | Finds the distance from the pole to their centre and adds it to and subtracts it from their radius. No justification that the pole is inside is needed. |
| | Greatest $5 + \sqrt{13}$, least $5 - \sqrt{13}$ | **A1** | Both, correctly identified. |
| | | **[4]** | |
| | **Total for Question 4** | **8** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **5(a)** | $x^2 + \left(y - \frac{\sqrt{3}}{2}\right)^2 = \ldots$ | **M1** | Completes the square in $y$, or compares with the general circle equation. Condone a slip in the constant. |
| | Centre $\left(0, \frac{\sqrt{3}}{2}\right)$, radius $\frac{\sqrt{3}}{2}$ | **A1** | Both, exact or awrt 0.866. |
| | $r^2\cos^2\theta + r^2\sin^2\theta = \sqrt{3}\,r\sin\theta$ | **M1** | Substitutes $x = r\cos\theta$, $y = r\sin\theta$, or uses $x^2 + y^2 = r^2$ with $y = r\sin\theta$. |
| | $r^2 = \sqrt{3}\,r\sin\theta$, $r \neq 0$<br>$\Rightarrow r = \sqrt{3}\sin\theta$ | **A1 AG** | Correct substituted equation leading without error to the printed result. Needs $r \neq 0$, or non-pole, stated when dividing by $r$. |
| | | **[4]** | |
| **5(b)** | $\sqrt{3}\sin\theta = 1 + \cos\theta$<br>$\Rightarrow \sqrt{3}\sin\theta - \cos\theta = R\sin(\theta - \alpha) = 1$ | **M1** | Equates the radii and reduces to one trigonometric function, e.g. any harmonic form with $R$ and $\alpha$ attempted, or half-angle identities. Condone sign slips. |
| | $R = 2,\ \alpha = \frac{\pi}{6}$ | **A1** | Or a correct equivalent single-function equation by another route. |
| | $\sin\left(\theta - \frac{\pi}{6}\right) = \frac{1}{2} \Rightarrow \theta = \ldots$ | **dM1** | Solves their equation for $\theta$ in $0 \leq \theta \leq \pi$, finding at least one value. |
| | $\theta = \frac{\pi}{3}$ ($\theta = \pi$ is the pole)<br>$P\left(\frac{3}{2}, \frac{\pi}{3}\right)$ | **A1** | cso. Exact. Needs $\theta = \pi$ rejected, or $\frac{\pi}{3}$ shown to be the only solution in $0 < \theta < \pi$, with one intersection stated. |
| | | **[4]** | |

*Question 5 continued*

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **5(c)** |  | | |
| | *Shape* | **B1** | Complete circle above the initial line, touching it at the pole. |
| | *Shape* | **B1** | Upper half of a cardioid from $(2,0)$, passing into $x < 0$ before reaching the pole; both curves labelled. |
| | *Intercepts* | **B1** | $P$ marked at the intersection, and $(2,0)$ and $(0,0)$ indicated. |
| | | **[3]** | |
| | **Total for Question 5** | **11** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **6(i)** | $r^4 = 5r^2\cos^2\theta + 4r^2\cos\theta\sin\theta + 2r^2\sin^2\theta$ | **M1** | Substitutes $x = r\cos\theta$, $y = r\sin\theta$ and $x^2 + y^2 = r^2$ throughout. Condone one slip. |
| | $r^2 = 5\cos^2\theta + 4\cos\theta\sin\theta + 2\sin^2\theta$ | **A1** | |
| | $r^2 = \frac{5}{2}(1 + \cos 2\theta) + 2\sin 2\theta + (1 - \cos 2\theta)$ | **M1** | Uses double-angle identities for all of $\cos^2\theta$, $\sin^2\theta$ and $\sin\theta\cos\theta$. Condone sign and coefficient slips. |
| | $r^2 = \dfrac{7 + 3\cos 2\theta + 4\sin 2\theta}{2}$ | **A1 AG** | Correct equation in $r$ and $\theta$ leading without error to the printed result; combined steps accepted. |
| | | **[4]** | |
| **6(ii)** | $3\cos 2\theta + 4\sin 2\theta = R\cos(2\theta - \alpha)$ $\dfrac{7 - R}{2} \leq r^2 \leq \dfrac{7 + R}{2}$ | **M1** | Expresses $3\cos 2\theta + 4\sin 2\theta$ as any single harmonic form and uses its bounds $\pm R$ in $r^2$. Condone a wrong $R$. |
| | $R = 5 \Rightarrow 1 \leq r^2 \leq 6$ | **A1** | |
| | Least distance 1, greatest distance $\sqrt{6}$ | **A1** | Both distances, exact. Not for the values of $r^2$ alone. |
| | | **[3]** | |
| | **Total for Question 6** | **7** | |

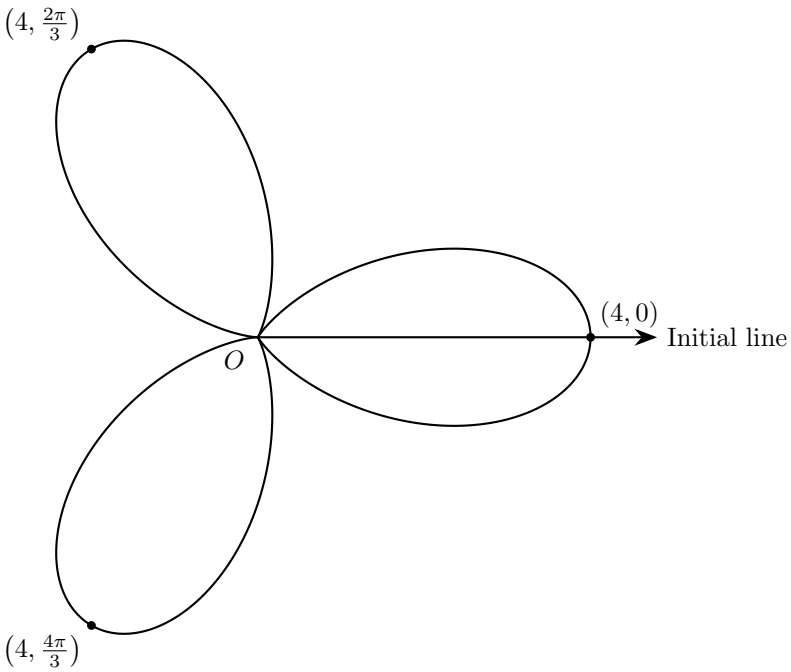
| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **7(a)** | $r\cos\left(\theta+\frac{\pi}{6}\right)=6$<br>$r\left(\cos\theta\cos\frac{\pi}{6}-\sin\theta\sin\frac{\pi}{6}\right)=6$ | **M1** | Multiplies through by $\cos\left(\theta+\frac{\pi}{6}\right)$ and expands the compound angle. Condone sign slips in the expansion. |
| | $\frac{\sqrt{3}}{2}x-\frac{1}{2}y=6$ | **dM1** | Replaces $r\cos\theta$ by $x$ and $r\sin\theta$ by $y$. |
| | $y=\sqrt{3}x-12$ | **A1** | cso. Must be in the form $y=mx+c$; another linear form such as $\sqrt{3}x-y=12$ scores A0. |
| | | **[3]** | |
| **7(b)** |  | | |
| | *Shape* | **B1** | Straight line with positive gradient crossing the initial line to the right of $O$, extending above and below it. |
| | *Half-line* | **B1** | Ray from $O$ below the initial line meeting $G_1$, with $\alpha$ marked between the ray and the initial line. |
| | *Labelled point* | **B1** | $\left(4\sqrt{3},0\right)$, $\left(6\sec\frac{\pi}{6},0\right)$ or awrt $(6.93,0)$, at the crossing of $G_1$ and the initial line. |
| | | **[3]** | |
| **7(c)** | $r=6\sec\left(-\alpha+\frac{\pi}{6}\right)$ | **M1** | Substitutes $\theta=-\alpha$ into the equation of $G_1$. |

*Question 7 continued*

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| | $\left(6\sec\left(\frac{\pi}{6}-\alpha\right),\ -\alpha\right)$ | **A1** | Both coordinates, oe such as $\frac{6}{\cos\left(\frac{\pi}{6}-\alpha\right)}$ or expanded. |
| | | **[2]** | |
| **7(d)** | $12\sqrt{3}=\frac{1}{2}\left(\text{“}4\sqrt{3}\text{”}\right)\left(\text{“}6\sec\left(\frac{\pi}{6}-\alpha\right)\text{”}\right)\sin\alpha$ | **M1 A1** | **M1** Forms the triangle's area in terms of $\alpha$, e.g. $\frac{1}{2}ab\sin C$ with their sides from $O$ and angle $\alpha$, or $\frac{1}{2}\int r^2\,\mathrm{d}\theta$, equal to $12\sqrt{3}$.<br>**A1** Correct equation, oe. |
| | $\sin\alpha=\cos\left(\frac{\pi}{6}-\alpha\right)$<br>$\quad=\frac{\sqrt{3}}{2}\cos\alpha+\frac{1}{2}\sin\alpha$<br>$\Rightarrow\tan\alpha=\ldots$ | **dM1** | Expands the compound angle and rearranges to an equation in $\tan\alpha$ alone. Condone sign slips. |
| | $\tan\alpha=\sqrt{3}\Rightarrow\alpha=\frac{\pi}{3}$ | **A1** | cso, with no other values given. |
| | | **[4]** | |
| | **Total for Question 7** | **12** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **8(i)** | $2(1+\cos 3\theta)=0 \Rightarrow \cos 3\theta=-1$<br>$\Rightarrow \theta=\ldots$ | **M1** | Sets $r=0$ and solves $\cos 3\theta=-1$ for at least one value of $\theta$. |
| | $\alpha=\frac{\pi}{3}$ | **A1** | In terms of $\pi$; 1.05 alone scores A0. |
| | | **[2]** | |
| **8(ii)(a)** | $f\left(\frac{2k\pi}{3}-\theta\right)=2[1+\cos(2k\pi-3\theta)]$<br>$=2[1+\cos 2k\pi\cos 3\theta+\sin 2k\pi\sin 3\theta]$ | **M1** | Expands $\cos(2k\pi-3\theta)$ by the addition formula, or uses the evenness and period $2\pi$ of cosine. |
| | $=2(1+\cos 3\theta)=f(\theta)$ | **A1** | Reaches $f(\theta)$ without error for general integer $k$, not a particular value. |
| | $f\left(\theta+\frac{2\pi}{3}\right)=2[1+\cos(3\theta+2\pi)]$<br>$=2[1+\cos 3\theta\cos 2\pi-\sin 3\theta\sin 2\pi]$ | **M1** | Expands $\cos(3\theta+2\pi)$ by the addition formula, or uses the period $2\pi$ of cosine. |
| | $=2(1+\cos 3\theta)=f(\theta)$ | **A1 AG** | Correct substituted expression leading without error to $f(\theta)$, with $=f(\theta)$ or a statement in words to conclude. |
| | | **[4]** | |
| **8(ii)(b)** | Reflection in $\theta=\beta$: $\theta\to 2\beta-\theta$<br>$2\beta=\frac{2k\pi}{3}\Rightarrow\theta=\frac{k\pi}{3}$ | **B1** | Lines $\theta=\frac{k\pi}{3}$ oe, linked to the first identity. |
| | $\theta=0,\ \frac{\pi}{3},\ \frac{2\pi}{3}$ | **B1** | All three and no others. |
| | Rotation through $\frac{2\pi}{3}$ | **B1** | |
| | | **[3]** | |

*Question 8 continued*

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **8(iii)** |  | | |
| | *Shape* | **B1** | Three equal closed lobes meeting at $O$, one centred on the initial line, the others in roughly the correct directions. |
| | $r_{\max} = 4$ | **B1** | |
| | $\cos 3\theta = 1 \Rightarrow 3\theta = 2k\pi \Rightarrow \theta = \ldots$ | **M1** | Sets $\cos 3\theta = 1$ and finds at least one nonzero value of $\theta$. |
| | $\theta = 0,\ \frac{2\pi}{3},\ \frac{4\pi}{3}$ | **A1** | All three, or 0, awrt 2.09 and 4.19, with no others in $0 \leq \theta < 2\pi$. |
| | | **[4]** | |
| | **Total for Question 8** | **13** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **9(a)** | $r\cos\left(\theta-\frac{\pi}{4}\right)=3$<br>$\Rightarrow \frac{r\cos\theta+r\sin\theta}{\sqrt{2}}=3$ | **M1** | Expands $\cos\left(\theta-\frac{\pi}{4}\right)$ and replaces $r\cos\theta$, $r\sin\theta$ by $x$, $y$. Condone a slip in the value of $\cos\frac{\pi}{4}$. |
| | $x+y=3\sqrt{2}$ | **A1 AG** | Correct expansion with $\cos\frac{\pi}{4}=\sin\frac{\pi}{4}=\frac{1}{\sqrt{2}}$, leading without error to the printed equation. |
| | | **[2]** | |
| **9(b)** | $\sin 2\theta = 2\cos^2\left(\theta-\frac{\pi}{4}\right)-1$<br>$\quad = \frac{18}{r^2}-1$<br>$r=5-2\left(\frac{18}{r^2}-1\right)$ | **M1** | Eliminates $\theta$ using $L$ and a double-angle identity, e.g. via $(x+y)^2=r^2(1+\sin 2\theta)$, to an equation in $r$ alone. Condone sign slips. |
| | $r^3-7r^2+36=0$ | **A1** | oe, e.g. $r=7-\frac{36}{r^2}$. |
| | $(r-3)(r-6)(r+2)=0\Rightarrow r=\ldots$ | **dM1** | Solves their cubic by any means, calculator included, to obtain values of $r$. |
| | $r=3,\ r=6$ ($r=-2$ rejected as $3\le r\le 7$) | **A1** | Both values, with $r=-2$ rejected for any valid reason. |
| | $\cos\left(\theta-\frac{\pi}{4}\right)=\frac{3}{\text{“}r\text{”}}$, $-\frac{\pi}{2}<\theta-\frac{\pi}{4}<\frac{\pi}{2}$<br>$\Rightarrow\theta=\ldots$ | **M1** | Substitutes a positive value of their $r$ into either equation and solves for $\theta$ in $-\frac{\pi}{4}<\theta<\frac{3\pi}{4}$. |
| | $\left(3,\frac{\pi}{4}\right),\ \left(6,-\frac{\pi}{12}\right),\ \left(6,\frac{7\pi}{12}\right)$<br>No other solutions in the range of $L$ | **A1** | cso. All three exact points and no others, with a statement that the cosine equations have no further solutions in the range of $L$. |
| | | **[6]** | |
| **Total for Question 9** | | **8** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **10(a)** | $(r\cos\theta)^2 - (r\sin\theta)^2 = 6r\cos\theta$ | **M1** | Substitutes both into the Cartesian equation. |
| | $r^2\cos 2\theta = 6r\cos\theta \Rightarrow r\cos 2\theta = 6\cos\theta$ | **A1** | |
| | $x > 0 \Rightarrow \cos\theta > 0$, $r > 0 \Rightarrow \cos 2\theta > 0$<br>$\Rightarrow -\frac{\pi}{4} < \theta < \frac{\pi}{4}$, $r = 6\cos\theta\sec 2\theta$ | **A1 AG** | Printed equation reached without error, with the range deduced from $\cos\theta > 0$ and $\cos 2\theta > 0$. A0 if the range is only stated. |
| | | **[3]** | |
| **10(b)(i)** | $0 < \cos 2\theta = \cos^2\theta - \sin^2\theta \leq \cos^2\theta \leq \cos\theta$ | **M1** | Attempts a valid justification: bounds $\cos 2\theta$ by $\cos\theta$ on the interval, sets $\frac{\mathrm{d}r}{\mathrm{d}\theta} = 0$, or writes $r^2 = 2x^2 - 6x$ for $x \geq 6$. |
| | $r = \dfrac{6\cos\theta}{\cos 2\theta} \geq 6$, equality only at $\theta = 0$ | **A1** | Or an equivalent conclusion: the stationary point shown to be a minimum, or $r^2$ increasing for $x \geq 6$. |
| | $N = (6, 0)$ | **A1** | |
| | | **[3]** | |
| **10(b)(ii)** | $6\sqrt{3}\cos 2\theta = 6\cos\theta$<br>$\Rightarrow \sqrt{3}(2\cos^2\theta - 1) = \cos\theta$ | **M1** | Equates $6\sqrt{3}$ to the polar equation and uses a double-angle formula to reach an equation in $\cos\theta$ only; or eliminates $y$ using $x^2 + y^2 = 108$. |
| | $2\sqrt{3}\cos^2\theta - \cos\theta - \sqrt{3} = 0$ | **A1** | oe, or $2x^2 - 6x - 108 = 0$ oe. |
| | $(2\cos\theta - \sqrt{3})(\sqrt{3}\cos\theta + 1) = 0 \Rightarrow \cos\theta = \ldots$ | **dM1** | Solves their three-term quadratic by factorising (product of constants correct), the formula or completing the square; calculator allowed. |
| | $\cos\theta = \frac{\sqrt{3}}{2}$ ($-\frac{1}{\sqrt{3}}$ rejected as $\cos\theta > 0$) | **A1** | Other root rejected with a reason, eg $\cos\theta > 0$ on the interval; or $x = 9$ with $x = -6$ rejected. |
| | $P\left(6\sqrt{3}, \frac{\pi}{6}\right)$, $Q\left(6\sqrt{3}, -\frac{\pi}{6}\right)$ | **A1** | Both points exact, angles in the stated interval; $\frac{11\pi}{6}$ for $Q$ scores A0. |
| | | **[5]** | |
| **10(b)(iii)** | $NP^2 = 6^2 + (6\sqrt{3})^2 - 2(6)(6\sqrt{3})\cos\frac{\pi}{6}$ | **M1** | Uses the cosine rule in triangle $ONP$ with $ON = 6$, their $OP$ from (b)(ii) and angle $\frac{\pi}{6}$, or the distance formula, or finds the Cartesian coordinates of $P$. |

*Question 10 continued*

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| | $NP = 6$ | **A1** | $NP = 6$, or $P(9, 3\sqrt{3})$ oe. |
| | $\angle ONP = \pi - 2\left(\frac{\pi}{6}\right)$ $[= \frac{2\pi}{3}]$<br>$\angle PNQ = 2\pi - 2\angle ONP = \ldots$ | **M1** | Uses $ON = NP$ to find $\angle ONP$ and symmetry about the initial line to reach $\angle PNQ$; or any valid route, eg the direction of $NP$. |
| | $\angle PNQ = \frac{2\pi}{3}$ | **A1** | Exact, in radians; 120° scores A0. |
| | | **[4]** | |
| | **Total for Question 10** | **15** | |